

Yang Zhou

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RESEARCH INTERESTS

Machine learning systems, networked systems

EDUCATION

Harvard University, Cambridge, MA, USA

Ph.D. in Computer Science

June 2024

M.S. in Computer Science

November 2021

Thesis: Network-Application Co-design for Efficient Datacenters

Advisors: Minlan Yu and James Mickens

Peking University, Beijing, China

B.S. in Computer Science

July 2018

Thesis: Towards Faster and More Accurate Data Stream Processing

Advisor: Tong Yang

EMPLOYMENT

University of California, Davis, Assistant Professor of Computer Science

July 2025 -

University of California, Berkeley, Postdoctoral researcher in Sky Computing Lab

July 2024 - now

Project: UCCL: an Efficient Collective Communication Library for GPUs

Supervisor: Ion Stoica

Google SRG and NetInfra, Research Intern

June 2021 - May 2023

VMware Research, Research Intern

July 2020 - September 2020

Meta/Facebook, Research Collaborator

November 2019 - May 2020

SenseTime, Software Engineering Intern

March 2018 - May 2018

PROFESSIONAL SERVICE

Leadership:

- Co-Chair: ACM SIGCOMM Artifact Evaluation 2024

Program Committees:

- USENIX NSDI: 2026
- USENIX OSDI: 2025
- ACM EuroSys: 2026
- ACM ASPLOS: 2026
- ACM SIGCOMM Workshop on eBPF and Kernel Extensions 2024, 2025
- ACM SIGCOMM Workshop on Networks for AI Computing 2025
- ACM SIGCOMM Poster/Demo 2023, 2024, 2025
- IEEE INFOCOM: Workshop on Networking Algorithms 2020

- UCCL, ultra and unified CCL for GPU communication, 560+ stars
<https://github.com/uccl-project/uccl>
- LEANN, the smallest vector index in the world for RAG, 2800+ stars
<https://github.com/yichuan-w/LEANN>
- NEO, an LLM inference engine built to save the GPU memory by CPU offloading
<https://github.com/NEO-MLSys25/NEO>
- eTran, implementing reliable network transports in the Linux kernel with eBPF
<https://github.com/eTran-NSDI25/eTran>
- DINT, running distributed transactions in the Linux kernel with eBPF
<https://github.com/DINT-NSDI24/DINT>
- Electrode, running Paxos consensus in the Linux kernel with eBPF
<https://github.com/Electrode-NSDI23/Electrode>

PUBLICATIONS

Preprints

- [1] Xingyu Xiang, Raj Joshi, Yuhan Liu, Jiayi Yao, Chenxingyu Zhao, Junchen Jiang, **Yang Zhou**, Eddie Kohler, Minlan Yu.
ShadowServe: Interference-Free KV Cache Fetching for Distributed Prefix Caching. [\[link\]](#)
Arxiv Sep 2025
- [2] Jiarong Xing, Yifan Qiao, Simon Mo, Xingqi Cui, Gur-Eyal Sela, **Yang Zhou**, Joseph Gonzalez, Ion Stoica.
Towards Efficient and Practical GPU Multitasking in the Era of LLM. [\[link\]](#)
Arxiv Aug 2025
- [3] Yichuan Wang, Shu Liu, Zhifei Li, Yongji Wu, Ziming Mao, Yilong Zhao, Xiao Yan, Zhiying Xu, **Yang Zhou**, Ion Stoica, Sewon Min, Matei Zaharia, Joseph E. Gonzalez.
LEANN: a Low-Storage Vector Index. [\[link\]](#)
Arxiv June 2025
- [4] **Yang Zhou**, Zhongjie Chen, Ziming Mao, ChonLam Lao, Shuo Yang, Pravein Govindan Kannan, Jiaqi Gao, Yilong Zhao, Yongji Wu, Kaichao You, Fengyuan Ren, Zhiying Xu, Costin Raiciu, Ion Stoica.
UCCL: an Efficient Collective Communication Library for GPUs. [\[link\]](#)
Arxiv April 2025
- [5] Shiyi Cao, Yichuan Wang, Ziming Mao, Pin-Lun Hsu, Liangsheng Yin, Tian Xia, Dacheng Li, Shu Liu, Yineng Zhang, **Yang Zhou**, Ying Sheng, Joseph Gonzalez, Ion Stoica.
Locality-Aware Fair Scheduling in LLM Serving. [\[link\]](#)
Arxiv Jan 2025
- [6] Yilong Zhao, Shuo Yang, Kan Zhu, Lianmin Zheng, Baris Kasikci, Yifan Qiao, **Yang Zhou**, Jiarong Xing, Ion Stoica.
BlendServe: Optimizing Offline Inference for Auto-regressive Large Models with Resource-aware Batch-ing. [\[link\]](#)
Arxiv Nov 2024
- [7] Yifan Qiao, Shu Anzai, Shan Yu, Haoran Ma, Shuo Yang, Yang Wang, Miryung Kim, Yongji Wu, **Yang Zhou**, Jiarong Xing, Joseph Gonzalez, Ion Stoica, Harry Xu.
ConServe: Harvesting GPUs for Low-Latency and High-Throughput Large Language Model Serving.

[\[link\]](#)

Arxiv Oct 2024

- [8] Shuo Yang, Ying Sheng, Yilong Zhao, Joseph Gonzalez, **Yang Zhou**, Ion Stoica, Lianmin Zheng.
Post-Training Sparse Attention with Double Sparsity. [\[link\]](#)
Arxiv Aug 2024

Conference

- [9] Xiangfeng Zhu, **Yang Zhou**, Yuyao Wang, Xiangyu Gao, Arvind Krishnamurthy, Sam Kumar, Ratul Mahajan, Danyang Zhuo. [\[Link\]](#)
Rethinking RPC Communication for Microservices-based Applications.
HotOS 2025.
- [10] Xuanlin Jiang, **Yang Zhou**, Shiyi Cao, Ion Stoica, Minlan Yu.
NEO: Saving GPU Memory Crisis with CPU Offloading for Online LLM Inference. [\[link\]](#)
MLSys 2025.
- [11] Zhongjie Chen, Qingkai Meng, ChonLam Lao, Yifan Liu, Fengyuan Ren, Minlan Yu, **Yang Zhou**.
eTran: Extensible Kernel Transport with eBPF. [\[link\]](#)
USENIX NSDI 2025.
- [12] **Yang Zhou**, Mark Wilkening, James Mickens, and Minlan Yu.
SmartNIC Security Isolation in the Cloud with S-NIC. [\[link\]](#)
ACM EuroSys 2024.
- [13] **Yang Zhou**, Xingyu Xiang, Matthew Kiley, Sowmya Dharanipragada, and Minlan Yu.
DINT: Fast In-Kernel Distributed Transactions with eBPF. [\[link\]](#)
USENIX NSDI 2024.
- [14] **Yang Zhou**, Zezhou Wang, Sowmya Dharanipragada, and Minlan Yu.
Electrode: Accelerating Distributed Protocols with eBPF. [\[link\]](#)
USENIX NSDI 2023.
- [15] **Yang Zhou**, Hassan Wassel, Sihang Liu, Jiaqi Gao, James Mickens, Minlan Yu, Chris Kennelly, Paul Turner, David Culler, Hank Levy, and Amin Vahdat.
Carbink: Fault-Tolerant Far Memory. [\[link\]](#)
USENIX OSDI 2022.
- [16] **Yang Zhou**, Ying Zhang, Minlan Yu, Guangyu Wang, Dexter Cao, Eric Sung, and Starsky Wong.
Evolvable Network Telemetry at Facebook. [\[link\]](#)
USENIX NSDI 2022.
- [17] **Yang Zhou**, Tong Yang, Jie Jiang, Bin Cui, Minlan Yu, Xiaoming Li, and Steve Uhlig.
Cold Filter: A Meta-Framework for Faster and More Accurate Stream. Processing [\[link\]](#)
ACM SIGMOD 2018.
- [18] Tong Yang, Jie Jiang, Peng Liu, Qun Huang, Junzhi Gong, **Yang Zhou**, Rui Miao, Xiaoming Li, and Steve Uhlig.
Elastic Sketch: Adaptive and Fast Network-Wide Measurements. [\[link\]](#)
ACM SIGCOMM 2018.
- [19] Omid Alipourfard, Masoud Moshref, **Yang Zhou**, Tong Yang, and Minlan Yu.
A Comparison of Performance and Accuracy of Measurement Algorithms in Software. [\[link\]](#)
ACM Symposium on SDN Research (SOSR) 2018.
- [20] Xiangyang Gou, Chenxingyu Zhao, Tong Yang, Lei Zou, **Yang Zhou**, Yibo Yan, Xiaoming Li, and Bin

Cui.

Single Hash: Use One Hash Function to Build Faster Hash Based Data Structures. [\[link\]](#)
IEEE International Conference on Big Data and Smart Computing (BigComp) 2018.

- [21] Tong Yang, **Yang Zhou**, Hao Jin, Shigang Chen, and Xiaoming Li.
Pyramid Sketch: A Sketch Framework for Frequency Estimation of Data Streams. [\[link\]](#)
VLDB 2017.
- [22] **Yang Zhou**, Peng Liu, Hao Jin, Tong Yang, Shoujiang Dang, and Xiaoming Li.
One Memory Access Sketch: A More Accurate and Faster Sketch for Per-Flow Measurement. [\[link\]](#)
IEEE Global Communications Conference (Globecom) 2017.
- [23] Junzhi Gong, Tong Yang, **Yang Zhou**, Dongsheng Yang, Shigang Chen, Bin Cui, and Xiaoming Li.
ABC: A Practicable Sketch Framework for Non-Uniform Multisets. [\[link\]](#)
IEEE International Conference on Big Data (BigData) 2017.

Workshop and Demo

- [24] **Yang Zhou**, Hao Jin, Peng Liu, Haowei Zhang, Tong Yang, and Xiaoming Li.
Accurate Per-Flow Measurement with Bloom Sketch. [\[link\]](#)
IEEE International Conference on Computer Communications Workshops (INFOCOM WKSHPS) 2018.

Journal

- [25] Zhuochen Fan, Gang Wen, Zhipeng Huang, **Yang Zhou**, Qiaobin Fu, Tong Yang, Alex X Liu, and Bin Cui.
On the Evolutionary of Bloom Filter False Positives - An Information Theoretical Approach to Optimizing Bloom Filter Parameters. [\[link\]](#)
IEEE Transactions on Knowledge & Data Engineering 2022.
- [26] Yuanpeng Li, Xiang Yu, Yilong Yang, **Yang Zhou**, Tong Yang, Zhuo Ma, and Shigang Chen.
Pyramid Family: Generic Frameworks for Accurate and Fast Flow Size Measurement. [\[link\]](#)
IEEE/ACM Transactions on Networking 2021.
- [27] Tong Yang, Jie Jiang, **Yang Zhou**, Long He, Jinyang Li, Bin Cui, Steve Uhlig, and Xiaoming Li.
Fast and Accurate Stream Processing by Filtering the Cold. [\[link\]](#)
The VLDB Journal 2019.
- [28] Tong Yang, Jie Jiang, Peng Liu, Qun Huang, Junzhi Gong, **Yang Zhou**, Rui Miao, Xiaoming Li, and Steve Uhlig.
Adaptive Measurements Using One Elastic Sketch. [\[link\]](#)
IEEE/ACM Transactions on Networking 2019.
- [29] **Yang Zhou**, Omid Alipourfard, Minlan Yu, and Tong Yang.
Accelerating Network Measurement in Software. [\[link\]](#)
ACM SIGCOMM Computer Communication Review 2018.

TALKS

-
- UCCL: an Efficient Collective Communication Library for GPUs
Meta, ByteDance (Seed), SJTU IPADS July 2025
ByteDance (Networking), NVIDIA June 2025
UC Berkeley SkyLab Summer Retreat, Broadcom May 2025
UC Berkeley SkyLab Winter Retreat January 2025
 - Network-Application Co-design for Efficient Datacenters
University of Toronto April 2024

NYU, Brown, UC Irvine, UWaterloo, UC Davis, Boston University	<i>March 2024</i>
UC Santa Cruz, University of Virginia, Purdue	<i>February 2024</i>
• Electrode: Accelerating Distributed Protocols with eBPF	
Duke University, ACE Center for Evolvable Computing, Google, USENIX NSDI	<i>April 2023</i>
Columbia University	<i>March 2023</i>
• Carbink: Fault-Tolerant Far Memory	
Cornell University	<i>November 2023</i>
WORDS workshop	<i>November 2022</i>
Microsoft Research Redmond, USENIX OSDI	<i>July 2022</i>
Google	<i>March & June 2022</i>
• Evolvable Network Telemetry at Facebook	
USENIX NSDI	<i>April 2022</i>
Boston University, Meta	<i>March 2022</i>
• Cold Filter: A Meta-Framework for Faster and More Accurate Stream Processing	
Harvard University	<i>October 2018</i>

STUDENTS

Current Students

- | | |
|---------------------|-------------------|
| • Shuang Ma (PhD) | <i>2025 - now</i> |
| • Yihan Zhang (PhD) | <i>2025 - now</i> |

Mentored Students

- | | |
|--|-------------------|
| • Zhongjie Chen, Tsinghua University PhD
Extensible kernel transport (NSDI 2025, [11]). | <i>2024 - now</i> |
| • Xuanlin Jiang, Peking University undergraduate → Harvard PhD
CPU offloading for online LLM inference (MLSys 2025, [10]). | <i>2024</i> |
| • Matt Kiley, Harvard College undergraduate → Clockwork Systems
Accelerating distributed transactions using eBPF (NSDI 2024, [13]). | <i>2023</i> |
| • Yunxi Shen, Tsinghua University undergraduate → Cornell PhD
Resource-efficient job scheduling in data centers. | <i>2023</i> |
| • Xingyu Xiang, Peking University undergraduate → Harvard PhD
Accelerating distributed transactions using eBPF (NSDI 2024, [13]). | <i>2023</i> |
| • Zezhou Wang, Peking University undergraduate → University of Washington PhD
Accelerating Paxos using eBPF (NSDI 2023, [14]). | <i>2022</i> |

TEACHING EXPERIENCE

- **Guest Lecture** on far memory, CS294-252: Architectures and Systems for Warehouse-Scale Computers, UC Berkeley *Nov 2023*
- **Teaching Assistant** for Prof. Minlan Yu, CS145: Networking at Scale, Harvard University *Spring 2021*
- **Teaching Assistant** for Prof. Tong Yang, Algorithm Design and Analysis, Peking University *Fall 2018*

PATENTS

- **Yang Zhou**, Hassan Wassel, Minlan Yu, Hank Levy, David Culler, and Amin Vahdat. “Fault Tolerant Disaggregated Memory”. Pending (US20230185666A1), December 2022.

AWARDS AND HONORS

- | | |
|---|------|
| • Google Ph.D. Fellowship in Systems and Networking | 2022 |
| • Finalist, Meta Ph.D. Fellowship in Networking | 2022 |
| • Graduate Fellowship, Harvard University | 2018 |
| • Excellent Bachelor Thesis (10/327), School of EECS, Peking University | 2018 |
| • New Academic Star Award (1/193), School of EECS, Peking University | 2018 |
| • Arawana Scholarship (2/193), Peking University | 2017 |
| • Pinyou Hudong Scholarship, School of EECS, Peking University | 2016 |
| • May Fourth Scholarship, Peking University | 2015 |